THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

MAIB7007 Accounting and Audit Automation

GENERAL INFORMATION

Instructor: Prof. Peeyush Taori
Email: peeyusht@hku.hk
Office: KK1201
Phone: 39171627
Consultation hours: By Appointment.

Tutor: TBD
Consultation hours: by Appointment.

Other important details:

COURSE DESCRIPTION

The primary focus of this course is to explore the integration and application of artificial intelligence (AI) technologies within the accounting profession. As the field of accounting continues to evolve with the rapid advancements in technology, AI has become a crucial component in streamlining and enhancing various accounting processes. This course aims to equip students with the necessary knowledge and skills to harness the potential of AI in the accounting landscape effectively. Throughout the course, students will gain insights into the fundamentals of artificial intelligence and its implications on accounting functions. By examining real-world case studies and engaging in hands-on exercises, students will develop a comprehensive understanding of how AI-driven tools can improve efficiency, reduce human error, and facilitate decision-making in the accounting profession.

Accounting and audit functions are being reshaped by two converging forces: automation, in the form of robotic process automation (RPA), workflow automation and intelligent document processing, and artificial intelligence (AI), in the form of machine learning, natural language processing and generative AI. This course examines how these technologies are combined to automate accounting and audit work end to end, from transaction capture, matching, reconciliation and period-end close, through to audit planning, risk assessment, evidence gathering, testing and reporting. Students will learn to view accounting and audit work as processes, identify which steps can be automated using rules and which require AI-enabled judgement, and design automated solutions with appropriate human oversight and controls.

The need for automation in accounting and audit has never been greater. Finance functions face growing transaction volumes, complex regulatory and reporting requirements, and pressure to deliver faster closes and real-time insights, while audit firms face expectations for full-population testing, continuous assurance and greater audit quality at lower cost. By automating repetitive and rules-based tasks such as invoice processing, journal entry posting, reconciliations and audit sampling, accountants and auditors can redirect effort towards analysis, professional judgement, risk assessment and advising. Accountants equipped with automation and AI competencies are therefore able to redesign processes, strengthen controls, improve fraud and anomaly detection, and lead digital transformation in their organizations.

Throughout this course, students will work hands-on with real-world use cases, including process mapping and automation opportunity assessment, automated bank and intercompany reconciliations, invoice-to-pay robots, intelligent document processing of invoices and contracts, AI-driven journal entry testing and audit risk scoring, continuous auditing and controls monitoring, and the use of generative AI assistants for audit documentation and analytical procedures. The curriculum also addresses the governance, control, ethical and professional-standards implications of automated and AI-enabled accounting and audit systems, including how to audit the robots and models themselves. By the end of the course, students will be able to design, build and evaluate automated accounting and audit solutions and contribute to the responsible transformation of the profession.

In addition to Excel usage, the course will introduce students to tools such as Alteryx for data automation, UiPath for robotic process automation and document understanding, MindBridge Ai Auditor for AI-driven audit analytics, and AnythingLLM and other generative AI tools for LLM-based assistants. These tools will be used to understand and apply techniques such as process mining and process mapping, RPA, optical character recognition and intelligent document processing, fuzzy matching, anomaly detection, classification, natural language processing, and continuous auditing. Students will be able to identify accounting and audit processes suitable for automation and build end-to-end solutions that combine automation and AI to deliver efficiency, accuracy and assurance.

COURSE OBJECTIVES

1. Understand the landscape of automation and AI technologies (RPA, intelligent document processing, machine learning and generative AI) in accounting and audit, and assess which processes are suitable for automation.
2. Design and build automated workflows and software robots for core accounting processes such as invoice processing, journal entries, reconciliations and period-end close.
3. Apply AI techniques such as anomaly detection, classification, NLP and generative AI to automate audit procedures including risk assessment, full-population testing and audit documentation.
4. Evaluate the governance, control, ethical and professional implications of automated and AI-enabled accounting and audit systems, and develop an automation roadmap for a finance or audit function.

PROGRAMME LEARNING OUTCOMES

PLO1: Acquiring and internalizing accounting and business knowledge and skills
PLO2: Applying and integrating accounting and business knowledge
PLO3: Inculcating professionalism and leadership
PLO4: Developing a global outlook

PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES

Course Learning Outcomes	Aligned Programme Learning Outcomes
CLO1: Explain the drivers, technologies and maturity levels of automation in accounting and audit, including RPA, intelligent automation, machine learning and generative AI	PLO 1, 2, 4
CLO2: Map accounting and audit processes, assess their suitability for automation, and quantify the expected benefits, costs and risks	PLO 1, 2
CLO3: Prepare, cleanse and integrate accounting data from ERP, general ledger and sub-ledger systems using repeatable data-automation workflows	PLO 1, 2
CLO4: Design and build software robots (RPA) to automate transactional accounting processes such as invoice-to-pay, reconciliations and journal entry posting	PLO 1, 2
CLO5: Apply intelligent document processing (OCR and NLP) to extract, validate and classify information from invoices, contracts and audit evidence	PLO 1, 2
CLO6: Automate audit procedures such as risk assessment, full-population testing, journal entry testing and sampling using AI-driven anomaly detection and risk scoring	PLO 1, 2, 5
CLO7: Design continuous auditing and continuous controls monitoring solutions for internal control, compliance and fraud detection	PLO 2, 5
CLO8: Use generative AI and LLM-based assistants for audit documentation, analytical procedures, contract review and financial reporting tasks, and critically evaluate their outputs	PLO 1, 2, 3

CLO9: Evaluate the governance, control, ethical and professional implications of automated and AI-enabled accounting and audit systems, including auditing the robots and models	PLO 1, 3, 4
CLO10: Develop and communicate an automation roadmap and business case for an accounting or audit function	PLO 2, 3, 5

COURSE TEACHING AND LEARNING ACTIVITIES

Course Teaching and Learning Activities		Study Load (% of study)
T&L1. Lectures with interactive presentations T&L2. Written assignment T&L3. Self-study and group discussions T&L4. Group project	30 20 40 40	
Total	130	100%

Assessment	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes
A1. Participation (In-class)	Attendance and discussion	15	CLO 1-10
A2. Assignment	Effort and accuracy	35	CLO 1-10
A3. Group Project	Effort, accuracy, analysis, and communications	50	CLO 1-10
	Total	100%	

STANDARDS FOR ASSESSMENT

Course Grade Descriptors

A+, A, A-	Demonstrate a strong understanding of all relevant knowledge •Handling questions professionally •High participation in discussions •Present arguments that have an element of originality •Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem solving skills •Excellent performance in assignments
B+, B, B-	•Demonstrate a good understanding of all relevant knowledge •Handling questions in a logical way •Good participation in discussions •Present arguments that go beyond the lecture and textbook •Achieve a standard of good performance in the exams with accurate computation and good analytical and problem solving skills •Good performance in assignments
C+, C, C-	•Demonstrate a basic understanding of the concepts involved •Fairly address questions as set •Some participation in discussions •Present arguments in a well-structure manner •Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem solving skills •Acceptable performance in assignments
D+, D	•Demonstrate a minimum understanding of the concepts involved •Barely address questions as set •Minimal or no participation in discussions •Present arguments in a marginally acceptable manner •Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem solving skills •Marginally acceptable performance in assignments
F	•Demonstrate a poor understanding of the concepts involved •Unable or unwilling to handle questions •Minimal or no participation in discussions •Present arguments poorly •Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem solving skills •Poorly performance in assignments

In-Class Group Work There will be several in-class Group Work for the groups. The purpose of in-class Group Work is to strengthen the concepts discussed in class and apply them in a practical setting.
Group Project Students in groups will have to work on a group project and submit a final report (totalling no more than 5,000 words).

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE	
Weeks	**Content**
1	
Week 1	**Foundations of Accounting and Audit Automation** The automation spectrum: RPA, workflow automation, intelligent automation, machine learning and generative AI Drivers of automation in accounting and audit: transaction volumes, regulation, continuous assurance and talent Mapping the accounting cycle (procure-to-pay, order-to-cash, record-to-report) and the audit process (planning, risk assessment, testing, reporting): where automation fits Process mapping and automation opportunity assessment: rules-based versus judgement-based tasks Automation governance, controls and ethics: designing human-in-the-loop processes Case Study: Map an order-to-cash process and prioritize automation candidates using a benefit-risk assessment.
2	
Week 2	**Data Automation for Accounting Processes** Data foundations for automation: ERP, general ledger and sub-ledger data, data quality and master data Building repeatable data pipelines with Alteryx: extraction, cleansing, joining and fuzzy matching Automating reconciliations (bank, intercompany, sub-ledger to GL) and period-end close tasks Automated exception reporting and monitoring dashboards for finance and audit teams Case Study: Build an automated bank reconciliation workflow with exception reporting in Alteryx.
3	
Week 3	**Robotic Process Automation and Intelligent Document Processing** RPA fundamentals with UiPath: recorders, variables, control flow and exception handling Automating invoice-to-pay, expense processing and journal entry posting Intelligent document processing: OCR, document understanding and NLP for invoices, contracts and audit evidence Combining RPA with AI (intelligent automation); bot governance: logging, access controls and bot audit trails Case Study: Build a UiPath robot that reads supplier invoices, validates them against purchase orders and posts the entries.
4	
Week 4	**AI-Enabled Audit Automation** Automating audit planning and risk assessment with AI: from sampling to full-population testing Anomaly detection, journal entry testing and transaction risk scoring with MindBridge Ai Auditor Continuous auditing and continuous controls monitoring; automated analytical procedures

	Generative AI and LLM assistants in audit: working papers, audit memos, contract and disclosure review (AnythingLLM); hallucination risk and verification Case Study: Use MindBridge to risk-score a general ledger, design follow-up audit procedures, and use an LLM assistant to draft the audit memo.
5	
Week 5	**Governance, Assurance and the Future of Automated Accounting and Audit** Auditing the automation: controls over robots, models and AI systems; model risk, explainability and bias Professional standards and regulatory expectations for technology-enabled accounting and audit Building an automation roadmap and business case; change management and the evolving role of accountants and auditors Emerging trends: agentic AI, autonomous close, real-time assurance and the skills of the future Group project presentations

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS

Recommend Textbook(s) or Suggested Reading(s) –:

1. Data Mining for Business Analytics: Concepts, Techniques, and Applications in R. Wiley. 2017. 1st edition.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

Conducting SFTL at the end of the course.

COURSE POLICY

1. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Academic dishonesty also includes, but is not limited to, providing false or misleading information to receive a postponement or an extension on an exam or other assignment.
2. An orderly learning environment is extremely important for this course. Disruptive behaviors are inconsiderate to other students as well as to the instructor, and are absolutely unacceptable. Talking during lectures, arriving to class late, and any other disruptions of mobile devices are not allowed; students who are responsible for any of these actions will be subject to academic penalty and will be asked to leave the classroom.
3. No late assignment submission will be accepted.
4. The instructor reserves all the rights to make necessary changes to the syllabus. If so, the changes will be announced as soon as possible.

ADDITIONAL COURSE INFORMATION

Software Tools Used in This Course

o Microsoft Excel, Alteryx, UiPath, MindBridge Ai Auditor, AnythingLLM

Tutorials

TBA